SUPPLY CHAIN & DELIVERY PERFORMANCE IMPROVEMENT

Business Analyst Project

Company: DataCo Global
Industry: E-commerce / Supply Chain
Role: Business Analyst
Project Type: Operational Improvement

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1. Business Background

DataCo Global is an e-commerce organization that manages a large volume of customer orders across multiple regions, product categories, and shipping methods. Efficient order fulfillment and timely delivery are important to maintaining customer satisfaction and controlling operational costs.

The organization has observed inconsistencies in delivery performance and wants to better understand the factors contributing to delayed orders.

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2. Business Problem

The organization lacks sufficient visibility into delivery performance across regions, shipping methods, product categories, and order periods.

Management needs to understand:

- Where delivery delays are occurring
- Which shipping methods are associated with poorer delivery performance
- Which regions experience higher levels of delays
- Which product categories may contribute to delivery issues
- Whether operational performance changes over time
- How delivery performance may affect business operations and costs

Without this analysis, management may find it difficult to identify operational bottlenecks and prioritize improvement initiatives.

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3. Business Impact

Delivery delays may result in:

- Customer dissatisfaction
- Increased customer complaints
- Higher operational costs
- Inefficient logistics planning
- Reduced customer retention

- Poor utilization of supply-chain resources

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4. Project Objective

Primary Objective

> **To analyze supply-chain and delivery performance, identify factors associated with delivery delays, and recommend process improvements that can improve on-time delivery and operational efficiency.**

Secondary Objectives

1. Measure overall delivery performance.
2. Identify areas with higher delivery delays.
3. Compare performance across shipping methods.
4. Analyze delivery performance across regions.
5. Identify product categories associated with delivery issues.
6. Analyze delivery trends over time.
7. Provide actionable recommendations to improve operational performance.

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5. Key Business Questions

Our analysis will answer:

1. What percentage of orders experience delivery delays?
2. Which regions have the highest delay rates?
3. Which shipping methods have the best and worst performance?
4. Which product categories experience more delivery issues?
5. How does delivery performance change over time?
6. Are certain operational patterns associated with delayed deliveries?
7. What actions could management take to improve delivery performance?

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6. Expected Outcome

The project will provide management with a data-driven understanding of delivery performance and the major areas requiring operational attention.

The analysis will be used to develop practical recommendations for improving delivery reliability, reducing delays, and supporting more efficient supply-chain operations.

2. Stakeholder Analysis

2.1 Stakeholder Identification

The following stakeholders are identified based on their involvement in order fulfillment, logistics, delivery operations, customer experience, financial management, and technology.

Stakeholder	Role / Responsibility	Interest	Influence
Operations Manager	Oversees overall order fulfillment and operational performance	High	High
Logistics Manager	Manages shipping, transportation and delivery operations	High	High
Warehouse Team	Handles order picking, packing and dispatch	High	Medium
Customer Service Team	Handles customer complaints and delivery-related issues	High	Medium
Finance Team	Monitors logistics and operational costs	Medium	High
IT Team	Supports systems, data and technology requirements	Medium	High
Customers	Receive products and experience the delivery process	High	Low

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2.2 Stakeholder Roles

Operations Manager

Primary decision-maker for operational improvements and project outcomes.

Logistics Manager

Provides expertise regarding shipping methods, carriers and delivery performance.

Warehouse Team

Provides information about order processing, packing and dispatch activities.

Customer Service Team

Provides insight into customer complaints and delivery-related issues.

Finance Team

Evaluates the financial impact of operational changes and shipping costs.

IT Team

Supports data availability, system requirements and potential technology changes.

Customers

The ultimate users affected by delivery performance and service quality.

2.3 Power / Interest Analysis

We'll categorize stakeholders using their level of **influence (Power)** and **interest**.

Category	Stakeholders	Engagement Strategy
High Power / High Interest	Operations Manager, Logistics Manager	Manage closely
High Power / Low Interest	Finance Team, IT Team	Keep satisfied
Low Power / High Interest	Warehouse Team, Customer Service Team	Keep informed
Low Power / Low Interest	Customers	Monitor / gather feedback

2.4 Stakeholder Communication Plan

Stakeholder	Information Needed	Communication Method	Frequency
Operations Manager	KPIs, findings, recommendations	Meetings / Reports	Weekly
Logistics Manager	Delivery performance and delay analysis	Meetings / Reports	Weekly
Warehouse Team	Process issues and improvement requirements	Team Meetings	Bi-weekly
Customer Service	Customer delivery issues	Meetings / Reports	Bi-weekly
Finance Team	Cost-related findings	Reports	Monthly
IT Team	Data and system requirements	Meetings	As required
Customers	Delivery experience	Surveys / Feedback	As required

2.5 Key Stakeholder

Primary Stakeholder: Operations Manager

The Operations Manager is considered the primary stakeholder because the project focuses on improving operational and delivery performance. The Operations Manager will use the analysis and recommendations to prioritize improvement initiatives.

3. Project Scope

3.1 In Scope

The following areas are included in the project:

- Order fulfillment and delivery performance
- Delivery delays and on-time delivery
- Shipping methods
- Regional delivery performance
- Product category performance
- Order volume and trends
- Shipping and logistics costs
- Identification of potential operational bottlenecks
- Analysis using Excel and SQL
- Development of business recommendations

3.2 Out of Scope

The following activities are outside the scope of this project:

- Changing or replacing logistics providers
- Negotiating supplier or carrier contracts
- Physical warehouse redesign
- Implementing new logistics technology
- Changes to employee staffing policies
- Customer compensation policies
- Actual implementation of recommended solutions

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4. Project Assumptions

The following assumptions will be used during the analysis:

1. The available dataset accurately represents the organization's historical order and delivery activity.
2. Required order, shipping and delivery information is available for analysis.
3. Delivery dates and shipping-related fields are sufficiently reliable for calculating delivery performance.
4. Historical patterns can be used to identify areas requiring further investigation.
5. Business stakeholders will validate the findings before implementing operational changes.

6. Recommendations are based on the available data and may require additional operational information before implementation.

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5. Project Constraints

The project has the following constraints:

1. Analysis is limited to the information available in the dataset.
2. The project does not have access to real-time delivery information.
3. External factors affecting delivery performance may not be captured in the dataset.
4. The analysis cannot independently confirm the operational root cause of every delay.
5. Recommendations are analytical and do not represent actual implemented business changes.
6. Data quality issues may affect the accuracy of some analysis.

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6. Project Success Criteria

The project will be considered successful if it:

- Establishes clear delivery-performance KPIs.
- Identifies major areas associated with delivery delays.
- Provides meaningful analysis by region, shipping method and product category.
- Identifies potential operational improvement areas.
- Produces clear and actionable recommendations.
- Provides stakeholders with sufficient information to support data-driven decision-making.

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7. Business Requirements

7.1 Requirement Identification

The requirements were developed based on the identified business problem, project objectives, stakeholder needs, and project scope.

Each requirement is assigned a unique ID so that it can be tracked throughout the project.

7.2 Business Requirements

ID	Business Requirement	Priority	Stakeholder
BR-001	The business should be able to measure overall delivery performance.	High	Operations Manager
BR-002	The business should be able to identify orders experiencing delivery delays.	High	Logistics Manager
BR-003	The business should be able to analyze delivery performance by region.	High	Operations Manager
BR-004	The business should be able to compare delivery performance across shipping methods.	High	Logistics Manager
BR-005	The business should be able to analyze delivery performance across product categories.	Medium	Operations Manager
BR-006	The business should be able to analyze delivery performance over time.	Medium	Operations Manager
BR-007	The business should be able to analyze shipping and logistics costs.	Medium	Finance Team
BR-008	The business should be able to identify areas that may require operational improvement.	High	Operations Manager

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7.3 Functional Requirements

These describe **what the analysis/solution should allow the business to do**.

ID	Functional Requirement	Priority
FR-001	Calculate the overall delivery delay rate.	High
FR-002	Calculate the number of delayed and on-time orders.	High
FR-003	Compare delivery performance across regions.	High
FR-004	Compare delivery performance across shipping methods.	High
FR-005	Analyze delivery performance by product category.	Medium

FR-006	Analyze delivery performance by order period.	Medium
FR-007	Calculate and compare relevant shipping-cost metrics.	Medium
FR-008	Allow identified performance issues to be supported by underlying data.	High

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7.4 Non-Functional Requirements

These describe **how the analysis/solution should perform or be presented**.

ID	Non-Functional Requirement	Priority
NFR-001	Analysis results should be accurate and based on validated data.	High
NFR-002	Business metrics should use clearly defined calculation methods.	High
NFR-003	Analysis should be understandable to non-technical business stakeholders.	High
NFR-004	Data should be handled consistently across Excel and SQL analysis.	High
NFR-005	Findings should be supported by measurable evidence.	High

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7.5 Requirement Priority Definition

We'll use a simple prioritization system:

High

Required to address the core business problem.

Medium

Provides additional business value but is not essential to the primary objective.

Low

Useful enhancement that can be considered later.

7.6 Business Success Metrics

The following metrics will be used to evaluate delivery performance:

- **On-Time Delivery Rate**
- **Delivery Delay Rate**
- **Number of Delayed Orders**
- **Average Shipping/Delivery Time**

- **Average Shipping Cost**
- **Delivery Performance by Region**
- **Delivery Performance by Shipping Method**

Note: Exact KPI definitions will be finalized after we inspect the dataset and validate the available fields.

8. AS-IS Process Analysis

8.1 Current-State Process

The current order fulfillment and delivery process can be represented as:

1. Customer Places Order
↓
2. Order is Recorded
↓
3. Order Processing
↓
4. Warehouse Picking & Packing
↓
5. Shipment Preparation
↓
6. Carrier / Shipping Handoff
↓
7. Transportation
↓
8. Customer Delivery
↓
9. Delivery Status Recorded

8.2 Process Description

Step	Process Activity	Responsible Team	Potential Issue
1	Customer places order	Customer / Sales	Incorrect order information
2	Order is recorded	Order Management	Data entry or system issues
3	Order is processed	Operations	Processing delays
4	Products are picked and packed	Warehouse	Inventory or staffing issues
5	Shipment is prepared	Warehouse / Logistics	Packing or dispatch delays
6	Shipment handed to carrier	Logistics	Carrier availability
7	Order transported	Carrier	Transit delays
8	Order delivered	Carrier	Delivery delays

9	Delivery status recorded	Operations / System	Incomplete or inaccurate status information

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8.3 Potential Pain Points

Based on the current-state process, the following areas require investigation:

1. Order Processing Delays

Orders may take longer than expected to move from order placement to fulfillment.

2. Warehouse Processing

Picking, packing or inventory-related issues may delay shipment preparation.

3. Shipping Handoff

Delays may occur between warehouse dispatch and carrier pickup.

4. Transportation

Transit times may vary depending on shipping method, region and other operational factors.

5. Delivery Performance

Some orders may not reach customers within the expected delivery timeframe.

6. Limited Performance Visibility

Without consistent analysis, management may have difficulty identifying where delays are concentrated.

8.4 AS-IS Analysis Objective

The purpose of analyzing the current-state process is to identify **where delays may occur and which process stages require further investigation**.

The actual causes of delays will be validated through the subsequent **Excel and SQL analysis** rather than assumed at this stage.

9. TO-BE Process Analysis

9.1 Proposed Future-State Process

The proposed process is:

1. Customer Places Order
↓
2. Order Validation
↓
3. Order Prioritization
↓
4. Warehouse Allocation
↓
5. Picking & Packing
↓
6. Carrier Assignment
↓
7. Shipment Tracking
↓
8. Delivery Monitoring
↓
9. Delivery Confirmation & Performance Recording

9.2 Future-State Process Description

Step	Proposed Activity	Expected Improvement
1	Customer places order	Order captured correctly
2	Order information is validated	Reduce incorrect/incomplete orders
3	Orders are prioritized based on relevant criteria	Improve processing efficiency
4	Orders are allocated to appropriate warehouse resources	Reduce fulfillment delays
5	Products are picked and packed	Improve dispatch readiness
6	Appropriate carrier/shipping method is assigned	Improve delivery performance
7	Shipment progress is monitored	Improve visibility
8	Delayed shipments are identified and escalated	Enable proactive intervention
9	Delivery outcome is recorded and reviewed	Support continuous improvement

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9.3 Proposed Improvements

1. Order Validation

Introduce validation of order information before processing to reduce errors that could create downstream delays.

2. Order Prioritization

Prioritize orders using defined business rules such as delivery commitments and operational requirements.

3. Improved Warehouse Coordination

Improve coordination between order processing and warehouse activities to reduce fulfillment delays.

4. Carrier / Shipping Method Evaluation

Use delivery-performance data to evaluate the effectiveness of different shipping methods.

5. Proactive Delay Monitoring

Identify potentially delayed shipments early so that operations teams can take corrective action.

6. Performance Measurement

Establish standardized delivery KPIs to continuously monitor operational performance.

9.4 AS-IS vs TO-BE Comparison

Add this table too:

Area	AS-IS	TO-BE
Order validation	Limited visibility	Standardized validation
Order processing	General processing flow	Prioritized processing
Warehouse coordination	Potential manual delays	Better allocation and coordination
Carrier selection	Performance may not be consistently evaluated	Data-driven evaluation
Delay management	Potentially reactive	Proactive monitoring
Performance measurement	Limited visibility	Standardized KPIs
Decision-making	Limited data-driven insight	Data-supported decision-making

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9.5 Expected Business Benefits

The proposed future state is expected to:

- Improve on-time delivery performance
- Reduce avoidable delivery delays
- Improve operational visibility
- Support better shipping-method decisions
- Improve coordination between operational teams
- Enable proactive identification of delivery issues
- Support data-driven decision-making

10. User Stories

10.1 User Story Format

We will use the standard Agile format:

> **As a [user], I want [function/capability], so that [business benefit].**

10.2 User Stories

ID	Related Requirement	User Story	Priority
US-001	BR-001	As an Operations Manager, I want to monitor overall delivery performance so that I can identify operational issues.	High
US-002	BR-002	As a Logistics Manager, I want to identify delayed orders so that I can investigate potential causes of delays.	High
US-003	BR-003	As an Operations Manager, I want to compare delivery performance across regions so that I can identify underperforming locations.	High
US-004	BR-004	As a Logistics Manager, I want to compare shipping methods so that I can identify methods associated with better delivery performance.	High
US-005	BR-005	As an Operations Manager, I want to analyze delivery performance by product category so that I can identify categories associated with delivery issues.	Medium
US-006	BR-006	As an Operations Manager, I want to analyze delivery performance over time so that I can identify trends and recurring issues.	Medium
US-007	BR-007	As a Finance Manager, I want to analyze shipping costs so that I can identify potential cost-efficiency opportunities.	Medium
US-008	BR-008	As an Operations Manager, I want to identify areas requiring operational improvement so that corrective actions can be prioritized.	High

10.3 Acceptance Criteria

We'll use **Given** → **When** → **Then**.

US-001 — Monitor Delivery Performance

Given valid order and delivery data is available
When the Operations Manager reviews the delivery analysis
Then overall delivery performance metrics should be available.

US-002 — Identify Delayed Orders

Given order and delivery information is available
When the Logistics Manager analyzes delivery records
Then delayed orders should be identifiable based on the defined delivery criteria.

US-003 — Compare Regional Performance

Given orders contain regional information
When the Operations Manager analyzes delivery performance by region
Then delivery performance should be comparable across regions.

US-004 — Compare Shipping Methods

Given orders contain shipping-method information
When the Logistics Manager compares shipping methods
Then delivery performance should be measurable for each shipping method.

US-005 — Analyze Product Categories

Given orders contain product-category information
When delivery performance is analyzed by category
Then the analysis should show performance differences between product categories.

US-006 — Analyze Trends

Given orders contain valid date information
When delivery performance is analyzed over time
Then trends and changes in delivery performance should be identifiable.

US-007 — Analyze Shipping Costs

Given shipping-cost information is available
When the Finance Manager reviews shipping costs
Then shipping costs should be measurable and comparable across relevant business dimensions.

US-008 — Identify Improvement Areas

Given delivery and operational analysis has been completed
When the Operations Manager reviews the findings
Then areas requiring further investigation or improvement should be clearly identified.

10.4 Definition of Done

We'll also establish a simple **Definition of Done** for our user stories.

A user story is considered **Done** when:

- The requirement has been analyzed.
- Relevant data has been validated.
- The required analysis has been completed.
- Results have been reviewed for accuracy.
- Findings are documented.
- The business requirement has been addressed.